PIERRE ELLIOTT TRUDEAU HIGH SCHOOL

SIGNIFICANT DIGITS, SCIENTIFIC NOTATION AND ROUNDING ERROR

Remember that a measurement is an approximation of an exact number.

Significant Digits (SD):

1. Non-zero digits are always significant, eg. 4.53 is 3 SD
2. Zeros at the beginning are never significant, eg. 0.00234 is 3 SD
3. Zeros between non-zeros are always significant, eg. 23.04 is 4 SD
4. Zeros at the end of an integer are usually not significant, eg. 4300 is 2 SD
5. Zeros following a decimal and behind a non-zero are significant, eg. 2.3500 is 5 SD

Examples:

Number	21.45 cm	60608 m	0.0029 s	0.01090 g	200. L
Number of SD's					

Rounding Off (RO):

1. If the first digit to be dropped is less than 5, leave the previous digit
   eg. 4.274 RO to 3 SD becomes 4.27
2. If the first digit to be dropped is more than 5, increase the previous digit by 1
   eg. 4.278 RO to 3 SD becomes 4.28
3. If the first digit dropped is 5 followed by non-zeros, increase the previous digit by 1
   eg. 7.273650001 RO to 5 SD becomes 7.2737

eg. 7.27365 (5SD) becomes 7.2736 eg. 7.275 (3 SD) becomes 7.28

Scientific Notation:

A number written as the product of a decimal number, equal to or greater than one and less than ten, and a power of 10, is said to be expressed in scientific notation,

Ex. 1) 32400 becomes _____ Ex. 2) 0.00324 becomes _____

The first factor (eg. 3.24) indicates the accuracy of the number and the second factor (eg. 10^4) indicates the magnitude of the number. All digits in the first factor are significant.

Addition and Subtraction of Measurements:

When measurements are added or subtracted, the measures should be in the same unit and the answer is to be rounded off to the least precise measurement (or the fewest number of digits to the right of the decimal).

Multiplication and Division of Measurements:

When measurements are multiplied or divided, then the number of significant digits in the answer must be the same as in the measure with the least number of significant digits.

Calculations with significant digits practice:

1) 357.89 + 0.002 =
2) 17.95 + 32.42 + 50 =
3) 5.5 +3.7 + 2.97 =
4) 3.24 x 5.6 =
5) 300 x 10.6 =
6) 0.059 x 6.95 =

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SPH3U PHYSICS
SIGNIFICANT DIGITS WORKSHEET #1

1. How many significant figures are there in each of the following?

a) 146.32 cm b) 1.8 cm c) 28.0 cm d) 28 cm

e) 700 m f) 700. m g) 1.00346 kg h) 0.00346 kg

i) 3.46×10^{-3} kg j) 3.00×10^{8} m/s k) 2 640 m/s l) 2.640 m/s

2. Round off the following to two significant digits:

a) 5.934 m b) 4.863 km c) 76.9 s

d) 0.3451 h e) 0.0181 m/s f) 0.0346 km/h

3. Express the following numbers in scientific notation with the proper number of significant digits:

a) 954 km b) 321.0 m c) 6 874 s

d) 6 400 J e) 6 400. m f) 0.000 64 min

g) 0.800 KJ h) 0.000 15 W i) 300 000 000 V

4. Express the following numbers in decimal notation:

a) 3.2×10^{5} m b) 3.2×10^{-5} m c) 3.2×10^{0} km

d) 8.791×10^{2} s e) 4.96×10^{-4} m f) 3×10^{8} s

5. Calculate the following expressing your answers with the proper number of significant figures:

a) $21.36\ g + 2.8\ g + 35.348\ g + 2.9164\ g$

b) $147.845\ m - 121.1\ m$

c) $54.612\ m - 3.0\ m$

d) the area of a rectangle whose length is 35.66 cm and whose width is 9.1 cm.

6. Calculate the values of the following expressing your answers in scientific notation with the proper number of significant digits:

a) $4.0 \times 10^5\ m/s \times 2.0 \times 10^3\ s$

b) $4.0 \times 10^5\ cm \div (2.0 \times 10^3\ s)$

c) $((2.0\ m^2)^3) \times 10^2$

d) $(5.0\ m/s \times 4.0\ s)^2 \times (1 \times 10^4\ N)$

e) $2.7 \times 10^3\ km \div (3.0 \times 10^{-4}\ h)$

f) $8.5 \times 10^3\ m - 4.5 \times 10^2\ m$

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SPH3U PHYSICS
SIGNIFICANT DIGITS WORKSHEET #2

1. How many significant figures are there in each of the following?

   a) 3.755 km b) 1000. cm c) 0.108504 mL d) 80.004 mg

   e) 0.000707 m f) 9.02×10^{8} kg g) 2.0×10^{1} m/s h) 5.34×10^{8} nm

2. Write each of the following in scientific notation with the correct number of significant digits.

   a) 40.0910 km/hr b) 0.003515 mL c) 38400. g d) 2.45 s

3. Express the following numbers in decimal notation:

   a) 9.033×10^{2} b) 4.7×10^{4} c) 5.0×10^{-8}

4. Evaluate the following expressing your answers with the proper number of significant figures:

   a) 0.907 cm + 4.5 cm – 3.007 cm b) 110 m/s x 4.77 s x 0.05 N/s

   c) 2.11×10^{-2} km – 9.002×10^{0} km d) 5.001×10^{5} mg + 3.87×10^{5} mg

   e) 589.44 ml ÷ 3.009 ml x 24 ml f) 37 s x 0.0208 m/s x 40.0 s ÷ 6.3 s

If interested there are short practice quizzes on significant digits in the website:

http://www.quia.com/tq/114241.html

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SPH3U PHYSICS
FORMULA MANIPULATION

A formula is an equation. Rearrange each equation for the term indicated. Show your steps. Always obey the equal sign when rearranging any equation. Whatever is done on the right side, should also be done on the left side.

a) $\overrightarrow{\Delta d} = \vec{v}\Delta t$ therefore $\vec{v}$ =

b) $\vec{v} = \dfrac{\overrightarrow{\Delta d}}{\Delta t}$ therefore Δt =

c) $\vec{v}_f = \vec{v}_i + \vec{a}\Delta t$ therefore $\vec{v}_i$ =

d) $\vec{v}_f = \vec{v}_i + \vec{a}\Delta t$ therefore Δt =

e) $\vec{a} = \dfrac{\overrightarrow{v_f} - \overrightarrow{v_i}}{\Delta t}$ therefore $\vec{v}_i$ =

f) $\vec{a} = \dfrac{\overrightarrow{v_f} - \overrightarrow{v_i}}{\Delta t}$ therefore $\vec{v}_f$ =

g) $\vec{a} = \dfrac{\overrightarrow{v_f} - \overrightarrow{v_i}}{\Delta t}$ therefore Δt =

h) $\vec{F}_{net} = m\vec{a}$ therefore $\vec{a}$ =

i) $\vec{a} = \dfrac{\overrightarrow{F_{net}}}{m}$ therefore m =

j) $F_g = \dfrac{GMm}{d^2}$ therefore d =

k) $F_g = \frac{GMm}{d^2}$ therefore m =

l) $R = \frac{\Delta V}{I}$ therefore ΔV =

m) $R = \frac{\Delta V}{I}$ therefore I =

n) $\overrightarrow{\Delta d} = \vec{v}_i \Delta t + {}^1/_2 \vec{a} \Delta t^2$ therefore $\vec{v}_i$ =

o) $\overrightarrow{\Delta d} = \vec{v}_i \Delta t + {}^1/_2 \vec{a} \Delta t^2$ therefore $\vec{a}$ =

p) $\overrightarrow{\Delta d} = \vec{v}_f \Delta t - {}^1/_2 \vec{a} \Delta t^2$ therefore Δt =

q) $E_T = mg\Delta h + {}^1/_2 mv^2$ therefore m =

r) $E_T = mg\Delta h + {}^1/_2 mv^2$ therefore v =

s) $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$ therefore f =

t) $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$ therefore d_o =

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SPH3U PHYSICS

Physics ⇔ Math Worksheet – Algebra and Substitution

Rearrange each equation for the term indicated. Show your steps. Whatever is done on the right side of the equal sign, should also be done on the left side.

1) $T = 2\pi\sqrt{\frac{L}{g}}$ (for g)

2) $m_1v_1 + m_2v_2 = m_1v_f + m_2v_f$ (for v_f)

3) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ (for R_2)

4) $m_1(x) = m_2(3 - x)$ (for x)

Evaluate the following using the given information. Rearrange the equation before substituting the values. In Physics, we substitute values with their units of measurements.

1. $\vec{v}_f = \vec{v}_i + \vec{a}\Delta t$ (Find $\vec{a}$, if $\vec{v}_i = 2$ m/s, $\vec{v}_f = 16$ m/s, $\Delta t = 2$ s)

2. $E_K = {}^1/_2mv^2$ (Find v, if $E_K = 10$ J, m = 4 kg)

3. $T = 2\pi\sqrt{\frac{L}{g}}$ (Find L, if T = 3 s, g = 50 m/s^2)

4. $\frac{\Delta V_p}{\Delta V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$ (Find N_p, if $N_s = 5$, $I_s = 12$ A , $I_p = 7.5$ A)

5. $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ (Find R_1, if $R_p = 8\ \Omega$, $R_2 = 16\ \Omega$)

6. $\overrightarrow{\Delta d} = \vec{v}_i\Delta t + {}^1/_2\vec{a}\Delta t^2$ (Find Δt, if $\vec{v}_i = 0$ m/s, $\overrightarrow{\Delta d} = 125$ m, $\vec{a} = 10$ m/s^2)

Solve the following word problems using the information and guided steps (I, II, III) provided.

1. If an airplane travels at 120 m/s (v), how long would it take (Δt) for the plane to travel a distance (Δd) of 300 meters?

   (I) List givens:
   v =
   Δd =
   Δt = ?

   Concept Equation: $v = \frac{\Delta d}{\Delta t}$

   (II) Derive Equation (isolate for Δt):

   (III) Substitute the given values into your derived equation and evaluate. [Always include units with their values. Round to the appropriate number of significant digits.]

2. A toy car accelerates from an initial velocity ($\overrightarrow{v_i}$) of 5 m/s, to a final velocity of ($\overrightarrow{v_f}$) of 17 m/s, in 6 seconds. Find the acceleration of the car.

   (I) List givens:
   $\overrightarrow{v_i}$ =
   $\overrightarrow{v_f}$ =
   Δt =
   $\vec{a}$ = ?

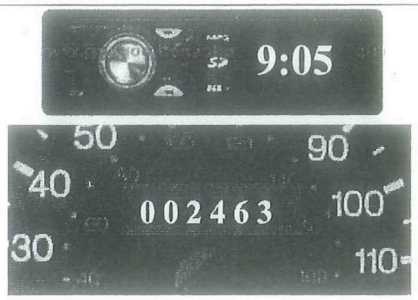
   Concept Equation: $\vec{v}_f = \vec{v}_i + \vec{a}\Delta t$

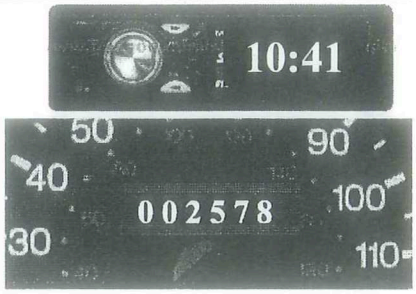
   (II) Derive Equation (solve for $\vec{a}$)

   (III) Substitute the given values into your derived equation for acceleration and evaluate.

A: The Physics Road Trip

You decide to take a trip to hear a lecture by one of your favourite physicists. When you begin driving, you glance at the clock in your car and also look at the odometer. As you pull in to the physics department parking lot, you look at the clock and the odometer a second time.





1. What is the readability of the clock and the odometer?

2. How much time did the trip take in **minutes**?

3. We would like to change this time interval into **seconds**. **Explain** how to do this.

4. Change the time interval into seconds. Show your math work. Use our guidelines to write the result. (Not a final result yet!)

5. What distance in kilometers did you travel?

6. We would like to change this distance into metres. **Explain** how to do this.

7. Change the distance into metres. Show your math work. Use our guidelines to write the result. (Not a final result yet!)

8. Calculate your average speed (average speed = distance traveled / time interval) during this trip twice – first, to get an answer with units of kilometers per minute, and again to get an answer with units of metres per second. Show your math work. Use our guidelines to record your **final** results.

kilometers per minute	**metres per second**

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THESE PREFIXES MAY BE APPLIED TO ALL SI UNITS

Multiples and Submultiples	Prefixes	Symbols
1 000 000 000 000 = 10^{12}	tera (tĕr'ȧ)	T
1 000 000 000 = 10^{9}	giga (j̃i'gȧ)	G
1 000 000 = 10^{6}	mega (mĕg'ȧ)	M
1 000 = 10^{3}	kilo (kĭl'ō)	k
100 = 10^{2}	hecto (hĕk'tō)	h
10 = 10	deka (dĕk'ȧ)	da
0.1 = 10^{-1}	deci (dĕs'ĭ)	d
0.01 = 10^{-2}	centi (sĕn'tĭ)	c
0.001 = 10^{-3}	mIlli (mĭl'ĭ)	m
0.000 001 = 10^{-6}	micro (mī'krō)	μ
0.000 000 001 = 10^{-9}	nano (năn'ō)	n
0.000 000 000 001 = 10^{-12}	pico (pē'kō)	p
0.000 000 000 000 001 = 10^{-15}	femto (fĕm'tō)	f
0.000 000 000 000 000 001 = 10^{-18}	atto (ăt'tō)	a

*Most commonly used

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FUNDAMENTAL UNITS & MKS

Quantity	Symbol	MKS Unit	Unit Abbreviation

UNIT CONVERSIONS

Conversion Factor:

A conversion factor is a quantity or ratio that relates two units of the same measurement and has a value of 1.

Conversion Factors for Metric Prefixes of Single Measurements:

In order to understand conversion factors for metric prefixes, one must be familiar with the staircase. In the case of the measurement length, with a base unit of metre (m), the staircase appears as:

km hm dam m dm cm mm

Each step of the staircase differs by a factor of 10.

To develop the idea of a conversion factor, consider an example for length:

We know that: *1 m = 100 cm*

Depending if we are converting from cm to m, or m to cm, we write the conversion factor for 'm' and 'cm' as:

_____ OR _____

In both cases the conversion factor relates the units of length – *m* to *cm* – and in both cases the conversion factor has a value of 1. **Because all conversion factors = 1, they do not changing the measurement's value, only its units.**

IMPORTANT: When setting up the conversion factor ensure that the units you wish to switch from divide out when multiplying by the conversion factor and you are left with the units you require.

EXAMPLE 1 (Length):

200 cm = _?_ *m*

To carry out the conversion we will multiply *200 cm* by a conversion factor that allows us to switch *cm* to *m*:

200 cm x

Conversion Factors for Quantities with Multiple Measurements:

Conversion factors become more important when dealing with unit conversions that have complicated or multiple conversions or more than one quantity of measurement that must be converted.

EXAMPLE 2 (Density):

1.2 g/cm^3 = _?_ *kg/m*^{3}

In this example we must convert mass in the numerator from *g* to *kg* and volume in the denominator from *cm*^{3} to *m*^{3}. As such we require two conversion factors:

Mass: _____ OR _____

Volume: To convert *cm*^{3} to *m*^{3}, we must use the fact that 100 cm = 1 m, but also that *(1 m)*^{3} *= (100 cm)*^{3} *or 1 m*^{3} *= 100*^{3} *cm*^{3}

Remember that the conversion factors are arranged in a way to cancel out the units that are given and leave the units that are required.

$$1.2\frac{g}{cm^3}\times$$

EXAMPLE 3 (Speed):

Using conversion factors ensures that the conversion is done accurately and efficiently.

Consider the example for **speed**: *34.5 km/h = ? m/s*

The conversion factors required are:

Length: _____

Time: *1 h = 60 min* and *1 min = 60 s* so... *1 h = 3600 s*

The conversion factor for hours to seconds, or seconds to hours is : _____ OR _____

Applying the conversion factors:

$$34.5\frac{km}{h}\times$$

EXAMPLE 4 (acceleration):

An example for **acceleration** is:

0.03 m/s^2 = ? km/h^2

The conversion factors are:

Length: _____ Time: _____

Applying the conversion factors:

$$0.03\frac{m}{s^2}\times$$

Note: When you see for example acceleration as *km/h/s* this can be written as $\frac{km}{h\times s}$

UNIT CONVERSIONS

Using conversion factors convert the following metric values into the units indicated in brackets. You must show the conversion factor(s) used (in brackets) and all of your steps. (Note that 1 m^3 = 1000 L)

1. a) (m) 323 mm

   b) (cm) 22 km

   c) (m^2) 43.2 cm^2

   d) (mm^2) 3.2 x 10^2 dam^2

   e) (mm^3) 3300 dm^3

   f) (cm^3) 5.8 x 10^{-2} m^3

   g) (L) 0.037 m^3

   h) (L) 249.7 cm^3

   i) (s) 2.5 min

   j) (min) 540 s

   k) (s) 2.5 h

   l) (h) 8500 s

2. a) (m/s) 60. km/h

   b) (km/h) 9 m/s

   c) (m/s) 120 km/h

   d) (km/h) 2.5 m/s

   e) (kg/m^3) 9.8 g/cm^3

   f) (g/cm^3) 5600 kg/m^3

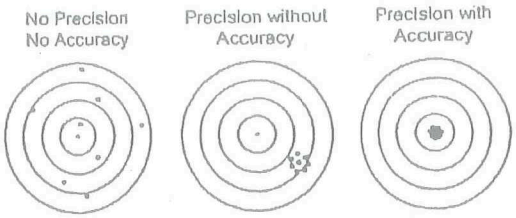
   g) (km/h/s) 2 m/s^2

   h) (m/s^2) 20 km/h/s

6 Error

6.1 Expressing Error in Measurement

In everyday usage, "*accuracy*" and "*precision*" are used interchangeably, but in science it is important to make a distinction between them. ***Accuracy*** refers to the closeness of a measurement to the accepted value for a specific quantity. ***Precision*** is the degree of agreement among several measurements that have been made in the same way. Think of shooting towards a bullseye target:



Error is the difference between an observed value (or the average of observed values) and the accepted value. The size of the error is an indication of the accuracy. Thus, the smaller the error, the greater the accuracy.

Every measurement made on every scale has some unavoidable possibility of error, usually assumed to be one-half of the smallest division marked on the scale. The accuracy of calculations involving measured quantities is often indicated by a statement of the ***possible error***. For example, you use a ruler calibrated in centimetres and millimetres to measure the length of a block of wood to be 12.6 mm (6 is the estimated digit in this measurement). The possible error in the measurement would be indicated by 12.6 ± 0.5 mm.

Relative error is expressed as a percentage, and is usually called *percentage error*.

'Absolute value' symbol means always positive

If multiple values, use the avg.

$$\text{percentage error} = \frac{|\text{experimental value} - \text{accepted value}|}{\text{accepted value}} \times 100\%$$

Same as "theoretical value"

Sometimes if two values of the same quantity are measured, it is useful to compare the precision of these values by calculating the ***percentage difference*** between them. If you have more than 2 measurements, use the maximum difference in measurements. + divide by the average of all values

$$\text{percentage difference} = \frac{|\text{measurement 1} - \text{measurement 2}|}{\left\{\frac{\text{measurement 1} + \text{measurement 2}}{2}\right\}} \times 100\%$$

OR if more than 2 values use.

$$\%\,\text{Diff} = \frac{|\text{Max} - \text{Min}|}{\text{Avg}} \times$$

★ MEMORIZE

Practice

1. Refer to page 3. State the possible error of measurements (a) to (c) in practice question #1.
2. At a certain location the acceleration due to gravity is 9.82 m/s^2[down]. Calculate the percentage error of the following experimental values of "*g*" at that location.
   (a) 8.94 m/s^2[down]
   (b) 9.95 m/s^2[down]
3. Calculate the percentage difference between the two experimental values (8.94 m/s^2 and 9.95 m/s^2) in question #2 above.

11 U Physics - Handout #2
Rounding and Percent Error/Difference

1. Round the following values to three significant digits:

a)	0.365839	b)	4.68163
c)	4.185	d)	73.50000
e)	8 935	f)	0.06525000

2. A jelly donut is measured to have a diameter of 6.25 cm. The accepted diameter is 6.43 cm. What is the percent error?

3. An acid is titrated with a base in chemistry class. A student performed the titration twice and found the concentration of the acid to be 4.75 M and 4.25 M. What is the percent difference between the two measurements?

For the following:

a) determine which would be appropriate to calculate: percent error or percent difference, justify your answer.

b) Provide a calculation for your answer in a)

4. On a test track, during trial runs, the maximum acceleration for a particular model of a new Porsche was found to be 4.91 m/s^2 and 5.23 m/s^2 .

5. In a physics experiment, the speed of sound was determined to be 362 m/s. In that particular situation, the actual value for the speed of sound would have been 340 m/s.

6. Using the same measuring device, a battery was tested twice, to find its voltage output. The first test gave a reading of 12.7 volts. The second test gave a reading of 11. 8 volts. The accepted voltage output for this particular battery is 12.0 volts.